Self-Supervised Latent Space Learning for Ultrasound Image Representation

General Info

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Project Abstract

This project builds a self-supervised latent space for ultrasound imaging, drawing on recent trends in foundation model representation learning, with the goal of exploiting the learned representations for unsupervised analysis such as clustering, structure discovery, and downstream clinical tasks.

Background and Motivation

Medical ultrasound is a widely used, non-invasive imaging modality, but its signals are inherently challenging: speckle noise, acquisition-dependent contrast, and high variability across patients and probe settings make supervised learning brittle and label-hungry. The lack of large annotated ultrasound datasets further limits the applicability of traditional deep learning pipelines.

A growing trend in medical imaging research addresses this through self-supervised representation learning — training models to construct compact, structured latent spaces from unlabeled data, without requiring manual annotation. These latent spaces, once learned, can be interrogated and exploited for a range of tasks: unsupervised clustering of tissue types, similarity-based retrieval, anomaly detection, and low-shot classification. This paradigm has shown strong results across modalities including MRI, pathology slides, and echocardiography, establishing that meaningful clinical structure is recoverable from the signal itself.

For ultrasound specifically, an open question is where in the signal processing chain — beamformed RF data or envelope-detected amplitude images — such a latent space is best constructed, and what structure it captures. The Stanford Ultrasound dataset, providing controlled acquisitions from tissue phantoms and in-vivo rat models, offers a well-characterized foundation for this investigation. This project aims to build such a latent space and systematically explore what it encodes about tissue properties and imaging conditions.

Student's Tasks Description

The student will complete the following tasks:

1. Data Preparation. Process the Stanford RF dataset to produce beamformed RF images and/or envelope-detected amplitude images. This includes implementing a preprocessing pipeline covering signal normalization, log-compression where applicable, and designing augmentation strategies appropriate for ultrasound signal statistics.

2. Latent Space Construction. Train a self-supervised model — drawing on established frameworks such as contrastive learning, masked prediction, or joint-embedding architectures — to learn a compact representation of the ultrasound signal. The student will make and justify design choices around architecture, input modality, and training objective.

3. Latent Space Analysis and Use. This is the core contribution. The student will systematically exploit the learned latent space through: dimensionality

reduction (UMAP, t-SNE) for visual structure discovery, unsupervised clustering (k-means, DBSCAN) to identify tissue or acquisition groupings, and linear probing to assess how well known labels (tissue type, phantom geometry, acquisition parameters) align with the learned structure.

4. Evaluation and Interpretation. Benchmark representation quality across signal stages and analyze which aspects of the signal — anatomical, physical, or artifactual — the model organizes in its latent space.

The student will gain expertise in medical signal processing, modern self-supervised learning, and systematic representation analysis methods.

Technical Prerequisites

- **Programming:** Proficiency in Python; hands-on experience with PyTorch.
- **Machine Learning:** Familiarity with deep learning fundamentals; basic knowledge of self-supervised or unsupervised learning is beneficial.
- **Signal/Image Processing:** Basic understanding of image processing concepts; prior exposure to ultrasound or medical imaging is a plus but not required.
- **Mathematics:** Linear algebra and probability at undergraduate level.
- **Tools:** Scientific Python stack (NumPy, SciPy, Matplotlib); version control with Git.

References

- Stanford Ultrasound Imaging & Instrumentation Lab. *RF Channel Data Datasets.* https://med.stanford.edu/ultrasound/resources/datasets.html
- Munim, A. et al. (2026). *EchoJEPA: A Latent Predictive Foundation Model for Echocardiography.* arXiv:2602.02603.
- He, K. et al. (2022). *Masked Autoencoders Are Scalable Vision Learners.* CVPR 2022.
- Chen, T. et al. (2020). *A Simple Framework for Contrastive Learning of Visual Representations.* ICML 2020.
- Caron, M. et al. (2021). *Emerging Properties in Self-Supervised Vision Transformers (DINO).* ICCV 2021.